A7q1 comsol

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1 Global Definitions

Date	Sep 20, 2025, 1:58:17 AM

GLOBAL SETTINGS

Name	A7q1 comsol.mph
Path	E:\Comsol codes\a7q1_comsol.mph
Version	COMSOL Multiphysics 6.0 (Build: 318)

USED PRODUCTS

COMSOL Multiphysics

COMPUTER INFORMATION

CPU	Intel64 Family 6 Model 142 Stepping 12, 2 cores
Operating system	Windows 10

1.1 PARAMETERS

PARAMETERS 1

Name	Expression	Value	Description
k	52[W/(m*K)]	52 W/(m·K)	
rho	7850[kg/m^3]	7850 kg/m³	
Cp	460[J/(kg*K)]	460 J/(kg·K)	
Q	0[W/m^3]	0 W/m³	

2 Component 1

2.1 DEFINITIONS

2.1.1 Coordinate Systems

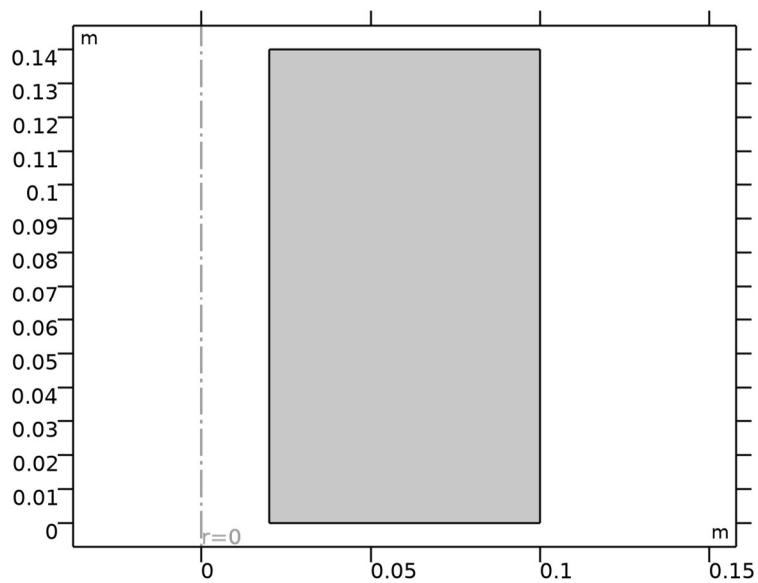
Boundary System 1

Coordinate system type	Boundary system
Tag	sys1

COORDINATE NAMES

First	Second	Third
t1	to	n

2.2 GEOMETRY 1

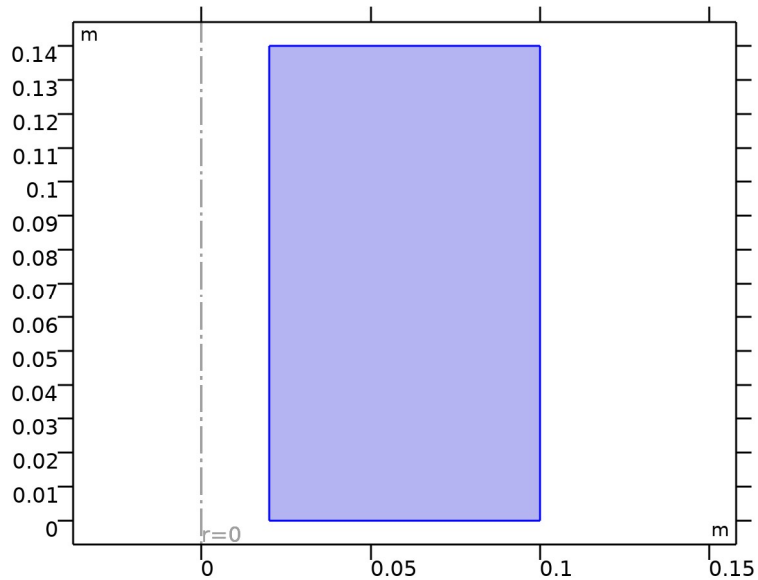


Geometry 1

UNITS

Length unit	m
Angular unit	deg

2.3 HEAT TRANSFER IN SOLIDS



Heat Transfer in Solids

EQUATIONS

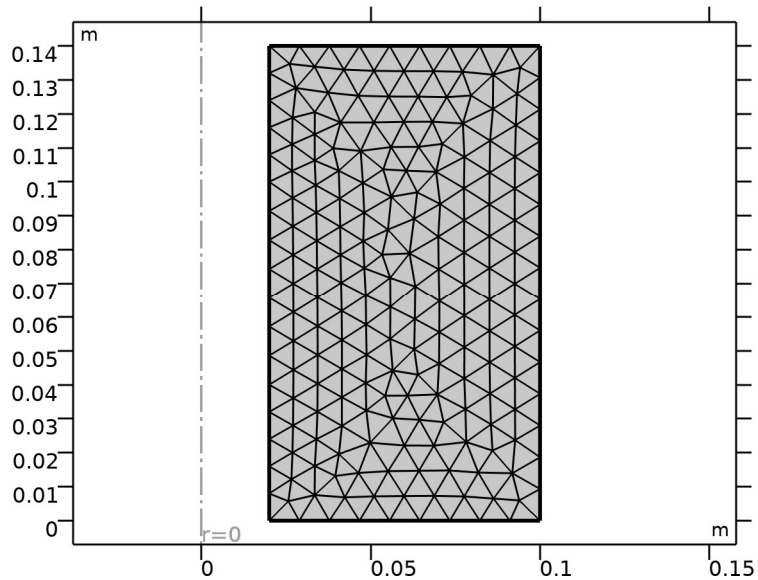
$$\rho C_p \mathbf{u} \cdot \nabla T + \nabla \cdot \mathbf{q} = Q + Q_{\text{ted}}$$

$$\mathbf{q} = -k \nabla T$$

FEATURES

Name	Level
Solid 1	Domain
Initial Values 1	Domain
Axial Symmetry 1	Boundary
Thermal Insulation 1	Boundary
Heat Flux 1	Boundary
Temperature 1	Boundary
Heat Source 1	Domain

2.4 MESH 1



Mesh 1

3 Study 1

COMPUTATION INFORMATION

Computation time	2 s

3.1 STATIONARY

STUDY SETTINGS

Description	Value
Include geometric nonlinearity	Off

PHYSICS AND VARIABLES SELECTION

Physics interface	Discretization
Heat Transfer in Solids (ht)	physics

MESH SELECTION

Geometry	Mesh
Geometry 1 (geom1)	mesh1

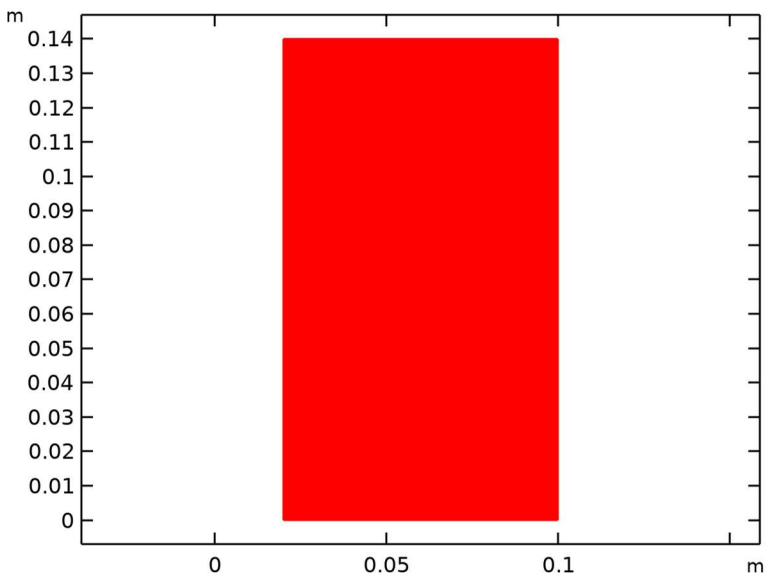
4 Results

4.1 DATASETS

4.1.1 Study 1/Solution 1

SOLUTION

Description	Value
Solution	Solution 1
Component	Component 1 (comp1)



Dataset: Study 1/Solution 1

4.1.2 Revolution 2D

DATA

Description	Value
Dataset	Study 1/Solution 1

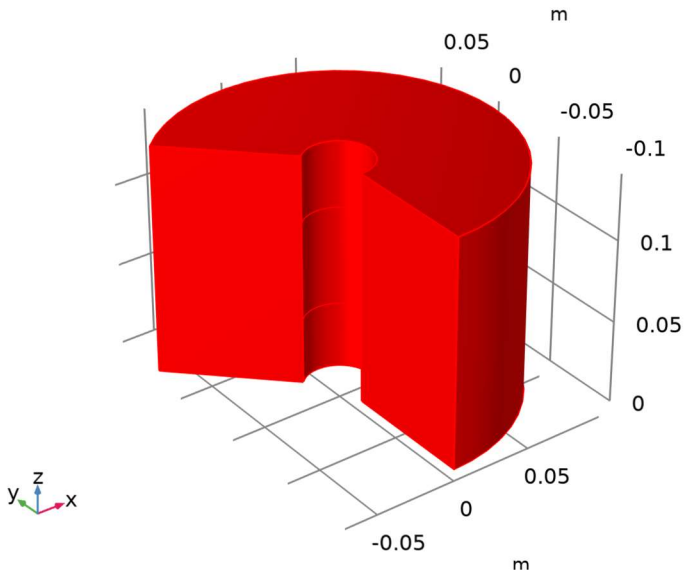
AXIS DATA

Description	Value
Axis entry method	Two points
Points	{{0, 0}, {0, 1}}

REVOLUTION LAYERS

Description	Value
Start angle	-90

Description	Value
Revolution angle	225



Dataset: Revolution 2D

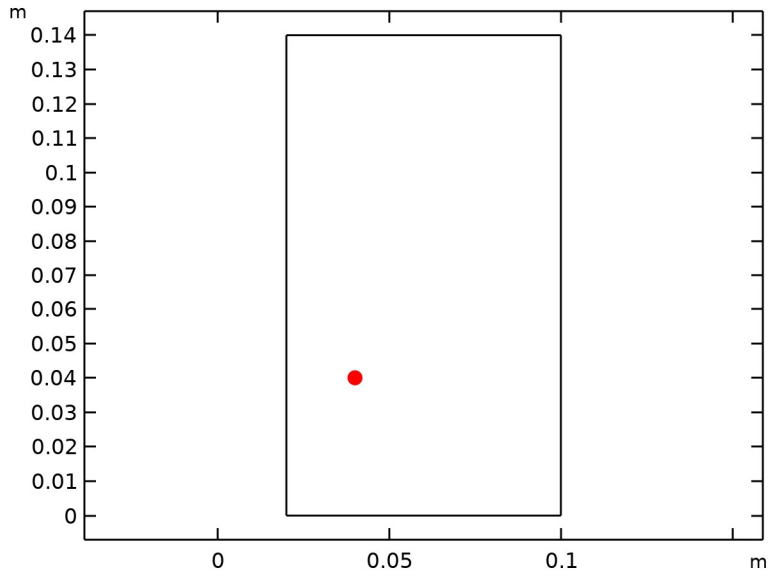
4.1.3 A

DATA

Description	Value
Dataset	Study 1/Solution 1

POINT DATA

Description	Value
Entry method	Coordinates
R	0.04
Z	0.04



Dataset: A

4.1.4 B

DATA

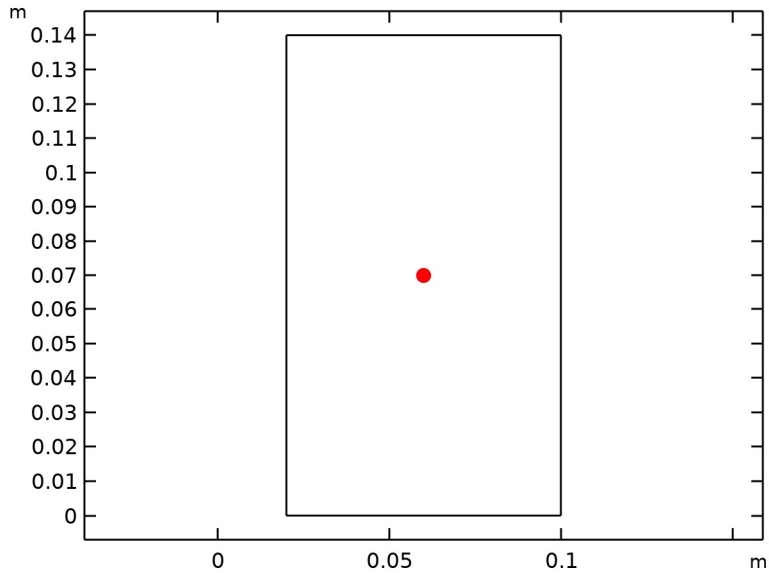
Description	Value
Dataset	Study 1/Solution 1

POINT DATA

Description	Value
Entry method	Coordinates
R	0.06
Z	0.07

ADVANCED

Description	Value
Point number variable	cpt1n



Dataset: B

4.1.5 C

DATA

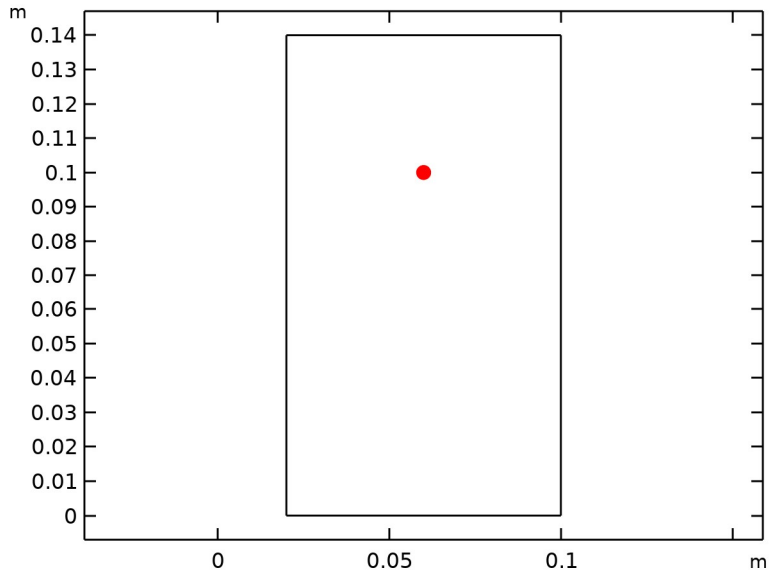
Description	Value
Dataset	Study 1/Solution 1

POINT DATA

Description	Value
Entry method	Coordinates
R	0.06
Z	0.1

ADVANCED

Description	Value
Point number variable	cpt1n



Dataset: C

4.1.6 R=0.03

DATA

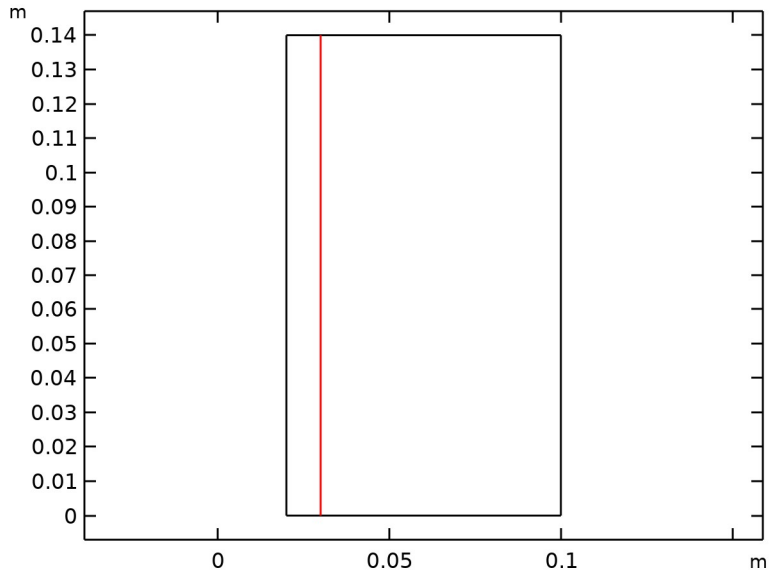
Description	Value
Dataset	Study 1/Solution 1

LINE DATA

Description	Value
Line entry method	Two points
Points	{{0.03, 0}, {0.03, 0.14}}

ADVANCED

Description	Value
Space variable	cln1x
Normal variables	{cln1nx, cln1ny}
Tangent variables	{cln1tx, cln1ty}



Dataset: R=0.03

4.1.7 R=0.05

DATA

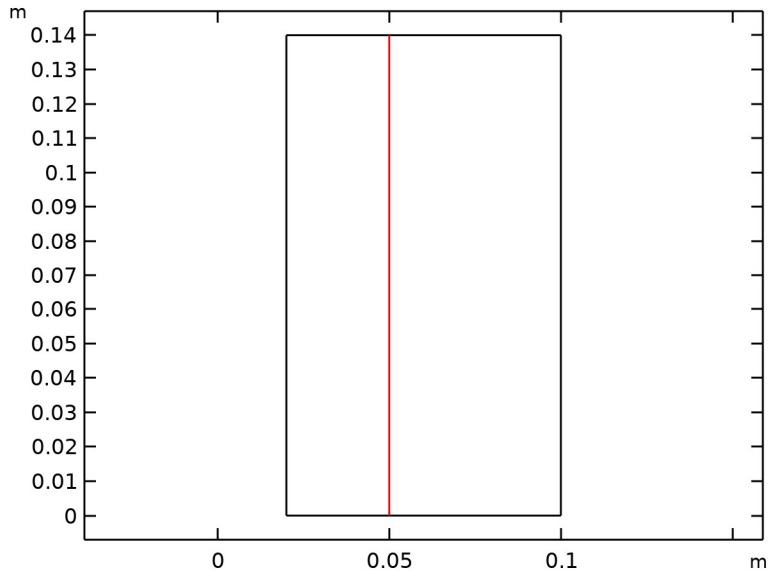
Description	Value
Dataset	Study 1/Solution 1

LINE DATA

Description	Value
Line entry method	Two points
Points	{{0.05, 0}, {0.05, 0.14}}

ADVANCED

Description	Value
Space variable	cln1x
Normal variables	{cln1nx, cln1ny}
Tangent variables	{cln1tx, cln1ty}



Dataset: R=0.05

4.1.8 R=0.07

DATA

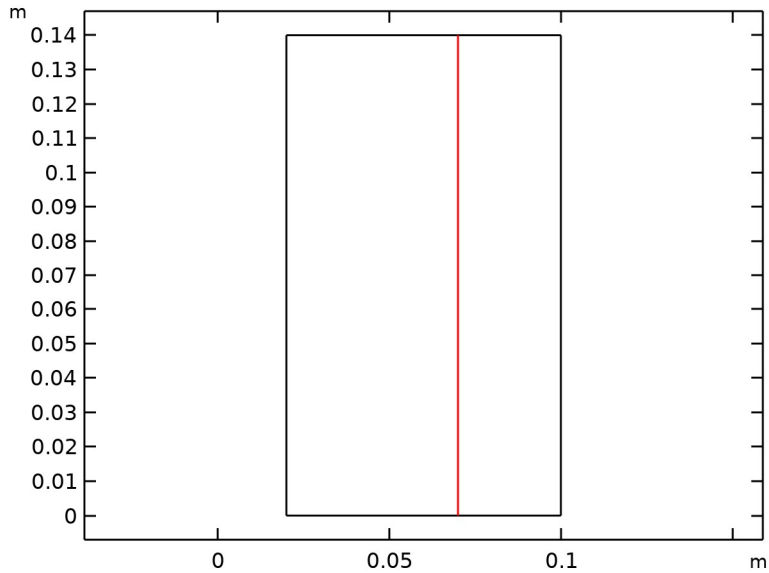
Description	Value
Dataset	Study 1/Solution 1

LINE DATA

Description	Value
Line entry method	Two points
Points	{{0.07, 0}, {0.07, 0.14}}

ADVANCED

Description	Value
Space variable	cln1x
Normal variables	{cln1nx, cln1ny}
Tangent variables	{cln1tx, cln1ty}



Dataset: R=0.07

4.1.9 Z=0.05

DATA

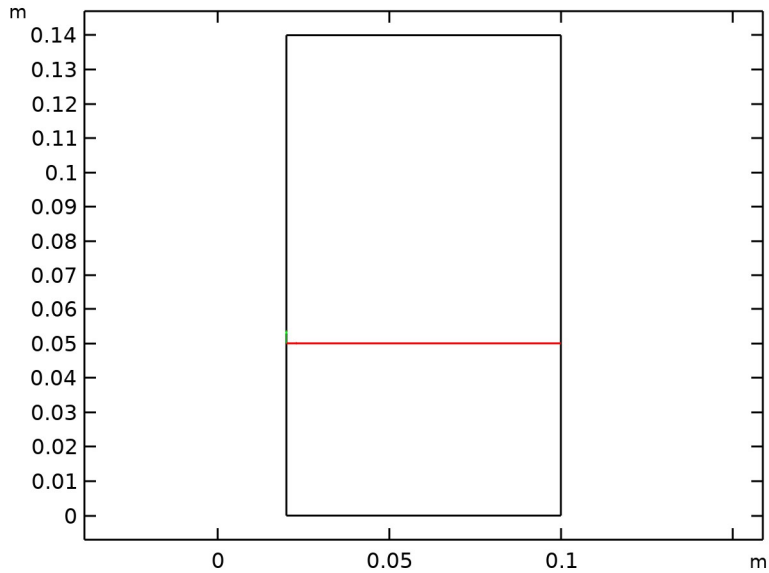
Description	Value
Dataset	Study 1/Solution 1

LINE DATA

Description	Value
Line entry method	Two points
Points	{{0.02, 0.05}, {0.1, 0.05}}

ADVANCED

Description	Value
Space variable	cln1x
Normal variables	{cln1nx, cln1ny}
Tangent variables	{cln1tx, cln1ty}



Dataset: Z=0.05

4.1.10 Z=0.08

DATA

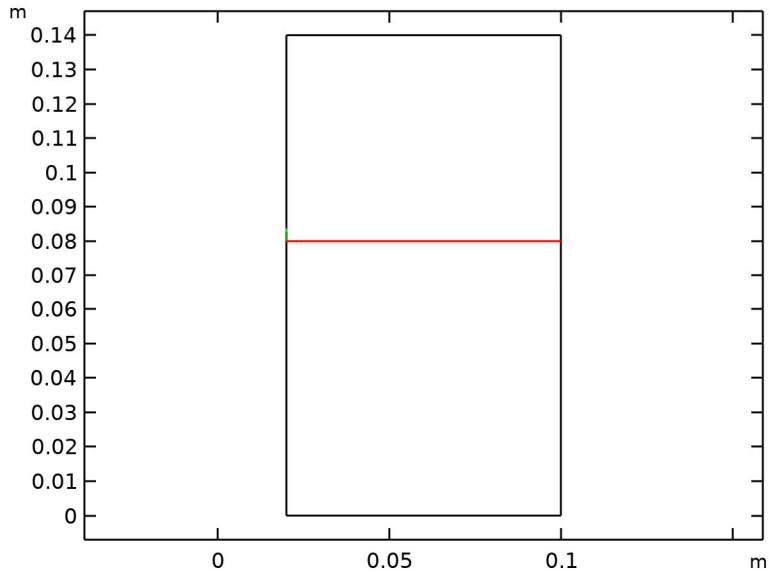
Description	Value
Dataset	Study 1/Solution 1

LINE DATA

Description	Value
Line entry method	Two points
Points	{{0.02, 0.08}, {0.1, 0.08}}

ADVANCED

Description	Value
Space variable	cln1x
Normal variables	{cln1nx, cln1ny}
Tangent variables	{cln1tx, cln1ty}



Dataset: Z=0.08

4.1.11 Z=0.1

DATA

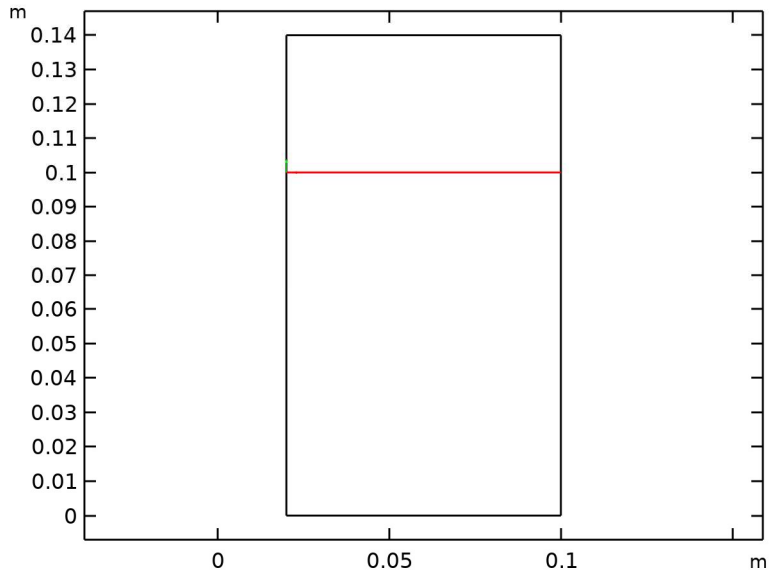
Description	Value
Dataset	Study 1/Solution 1

LINE DATA

Description	Value
Line entry method	Two points
Points	{{0.02, 0.1}, {0.1, 0.1}}

ADVANCED

Description	Value
Space variable	cln1x
Normal variables	{cln1nx, cln1ny}
Tangent variables	{cln1tx, cln1ty}



Dataset: Z=0.1

4.2 DERIVED VALUES

4.2.1 Point Evaluation 1

OUTPUT

Evaluated in	Table 1

DATA

Description	Value
Dataset	A

EXPRESSIONS

Expression	Unit	Description
T	K	Temperature

4.2.2 Point Evaluation 2

OUTPUT

Evaluated in	Table 2

DATA

Description	Value
Dataset	B

EXPRESSIONS

Expression	Unit	Description

Expression	Unit	Description
T	K	Temperature

4.2.3 Point Evaluation 3

OUTPUT

Evaluated in	Table 3

DATA

Description	Value
Dataset	C

EXPRESSIONS

Expression	Unit	Description
T	K	Temperature

4.3 TABLES

4.3.1 Table 1

Point Evaluation 1

Temperature (K), Point: (0.04, 0.04)
332.96

4.3.2 Table 2

Point Evaluation 2

Temperature (K), Point: (0.06, 0.07)
313.10

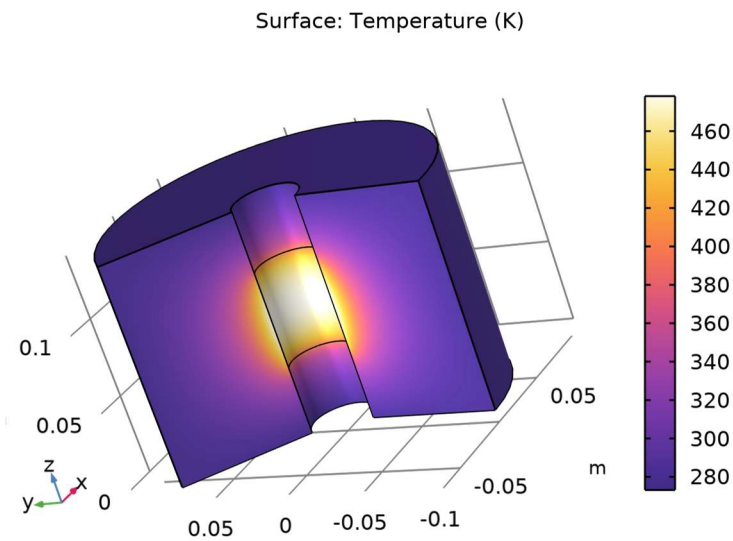
4.3.3 Table 3

Point Evaluation 3

Temperature (K), Point: (0.06, 0.1)
302.40

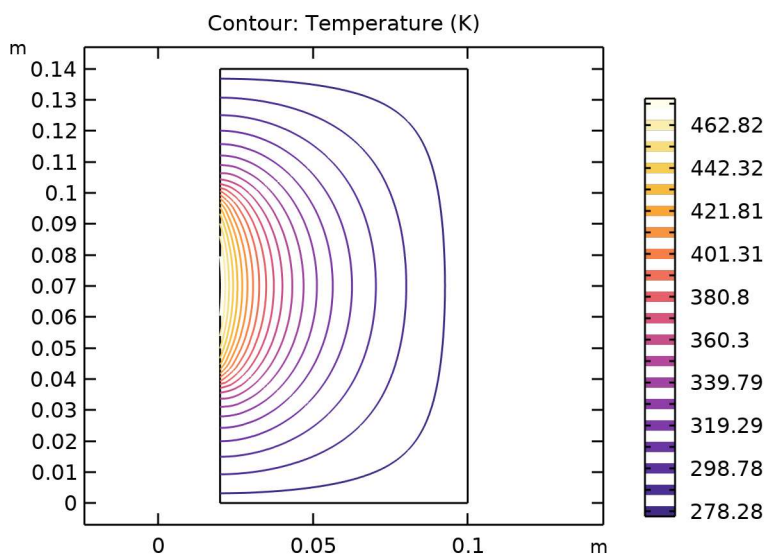
4.4 PLOT GROUPS

4.4.1 Temperature, 3D (ht)



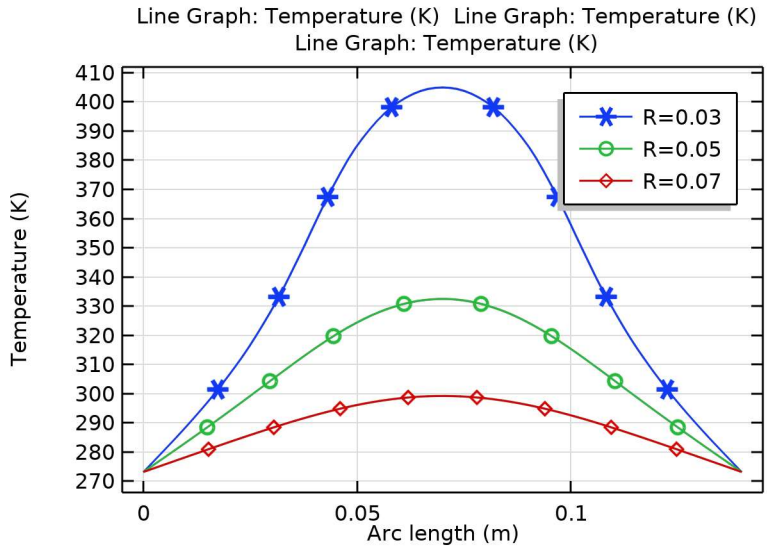
Surface: Temperature (K)

4.4.2 Isothermal Contours (ht)



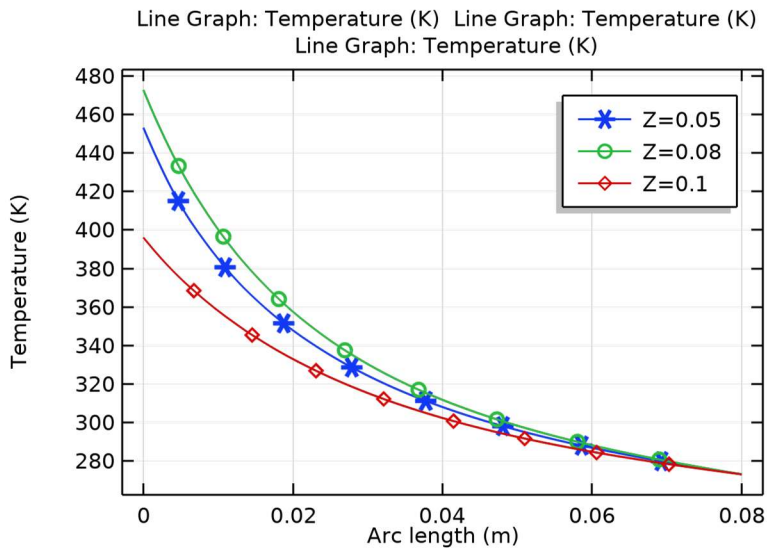
Contour: Temperature (K)

4.4.3 Temp profile(Radial)



Line Graph: Temperature (K) Line Graph: Temperature (K) Line Graph: Temperature (K)

4.4.4 Temp profile(Axial)



Line Graph: Temperature (K) Line Graph: Temperature (K) Line Graph: Temperature (K)